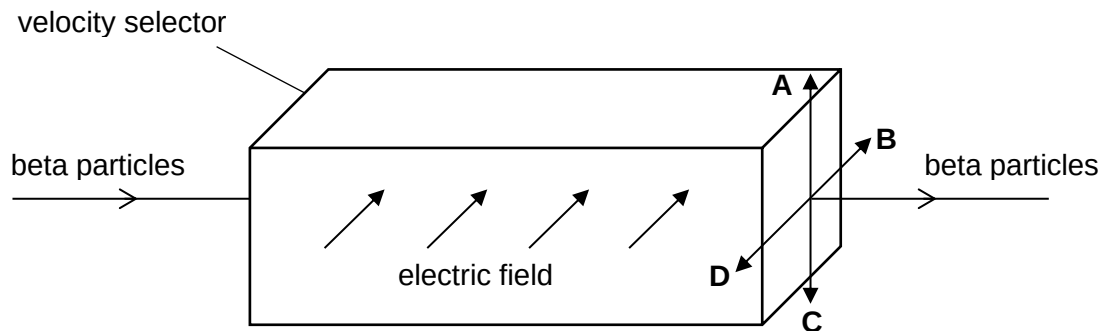


25 A beam of beta particles enters a velocity selector. An electric field is applied in a horizontal direction, perpendicular to the beam of beta particles, as shown in the diagram below.

A magnetic field is applied perpendicular to the beam such that beta particles of a particular speed leave the selector undeflected.

In which direction is the magnetic field?



Answer: A

Beta particles are negatively charged.

Electric force acts out of the page, opposite to the electric field.

To remain undeflected, the magnetic force must be equal in magnitude and opposite direction to the electric force, so it must act into the page.

By FLHR, magnetic field acts upwards.

26 A conner disc of radius r rotates about its centre O at a constant speed. It is placed in a